Duty at home

DM11 · One face and its height · 50 min.

In an orthonormal coordinate system of origin O, the points are considered $A(2;0;0)$, $B(0;2;0)$ and $C(0;0;2)$.

1. Demonstrate that the triangle ABC is equilateral and calculate its area.

2. Show that the vector $\vec{n} = (1;1;1)$ is normal on the plane (ABC).

3. Note H the orthogonal project of O on (ABC). Justify that its coordinates are of the form $(t;t;t)$. Determine t using $\overrightarrow{AH} \cdot \vec{n} = 0$.

4. Calculate OH, then the volume of tetrahedron $OABC$ by taking ABC for base. Check this volume by taking OAB for the base.